

Preoperative Assessment and Premedication MCQ – Chapter 4

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 3: Quintessential in Anesthesia (Chapter 4)

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. What is the primary purpose of preoperative assessment?

- A. To delay the surgery
- B. To evaluate the patient's fitness for anesthesia and surgery
- C. To prescribe antibiotics
- D. To schedule follow-up visits

Ans: B

Q2. Which ASA physical status classification denotes a normal healthy patient?

- A. ASA I
- B. ASA II
- C. ASA III
- D. ASA IV

Ans: A

Q3. ASA II classification refers to a patient with:

- A. Severe systemic disease that limits activity
- B. Mild systemic disease with no functional limitation
- C. Incapacitating disease that is a constant threat to life
- D. A moribund patient not expected to survive 24 hours

Ans: B

Q4. Which ASA classification is given to a moribund patient not expected to survive 24 hours with or without surgery?

- A. ASA III
- B. ASA IV
- C. ASA V
- D. ASA VI

Ans: C

Q5. The suffix 'E' in ASA classification denotes:

- A. Elderly patient
- B. Emergency surgery
- C. Endocrine disorder
- D. Extremity surgery

Ans: B

Q6. During preoperative assessment, Mallampati classification is used to assess:

- A. Cardiac function
- B. Airway and ease of intubation
- C. Renal function
- D. Respiratory reserve

Ans: B

Q7. Mallampati Class I visualization includes:

- A. Soft palate only
- B. Soft palate, fauces, uvula, and anterior and posterior tonsillar pillars
- C. Soft palate and base of uvula
- D. Hard palate only

Ans: B

Q8. Mallampati Class IV indicates visualization of: A. Full uvula and tonsillar pillars B. Soft palate and base of uvula C. Hard palate only (soft palate not visible) D. Soft palate, fauces, and uvula	Ans: C
Q9. Which of the following is NOT a component of the preoperative airway assessment? A. Thyromental distance B. Mallampati score C. Neck mobility D. Ankle-brachial index	Ans: D
Q10. A thyromental distance of less than 6 cm suggests: A. Easy intubation B. Difficult intubation C. Normal airway D. Need for tracheostomy	Ans: B
Q11. Preoperative fasting guidelines recommend NPO for clear fluids for how many hours before elective surgery in adults? A. 1 hour B. 2 hours C. 4 hours D. 6 hours	Ans: B
Q12. The recommended fasting time for solid food before elective surgery in adults is: A. 2 hours B. 4 hours C. 6 hours D. 8 hours	Ans: C
Q13. Which investigation is mandatory in all patients above 40 years of age undergoing surgery? A. CT scan of brain B. ECG C. Echocardiography D. Pulmonary function test	Ans: B
Q14. Which of the following drugs should be continued on the morning of surgery? A. Oral hypoglycemics B. Antihypertensive medications C. Warfarin D. Aspirin (for minor surgery)	Ans: B

Q15. Preoperative instructions include stopping aspirin how many days before elective surgery? A. 1 day B. 3 days C. 7 days D. 14 days	Ans: C
Q16. The primary goal of premedication is: A. To induce general anesthesia B. To allay anxiety and facilitate smooth induction C. To replace intraoperative anesthetics D. To provide postoperative analgesia only	Ans: B
Q17. Which of the following is a desired effect of premedication? A. Increased salivation B. Anxiolysis C. Hypertension D. Bronchospasm	Ans: B
Q18. Which benzodiazepine is most commonly used for premedication? A. Chlordiazepoxide B. Midazolam C. Flurazepam D. Clonazepam	Ans: B
Q19. The antisialagogue effect of premedication is achieved by using: A. Midazolam B. Glycopyrrolate C. Fentanyl D. Ondansetron	Ans: B
Q20. Which anticholinergic drug crosses the blood-brain barrier and can cause central effects? A. Glycopyrrolate B. Atropine C. Ipratropium D. Hyoscine butylbromide	Ans: B
Q21. Which of the following is NOT a goal of premedication? A. Reduction of gastric volume and acidity B. Prevention of autonomic reflexes C. Induction of surgical anesthesia D. Reduction of postoperative nausea and vomiting	Ans: C

Q22. Ranitidine given as premedication acts by: A. Blocking muscarinic receptors B. Blocking H2 receptors to reduce gastric acid secretion C. Blocking dopamine receptors D. Enhancing gastric motility	Ans: B
Q23. Metoclopramide given as premedication helps by: A. Reducing anxiety B. Increasing gastric emptying and increasing lower esophageal sphincter tone C. Providing analgesia D. Reducing bronchial secretions	Ans: B
Q24. Sodium citrate given preoperatively is used to: A. Reduce bronchial secretions B. Neutralize gastric acidity C. Provide sedation D. Prevent bradycardia	Ans: B
Q25. Which opioid is commonly used as premedication for its analgesic and sedative properties? A. Naloxone B. Morphine C. Naltrexone D. Flumazenil	Ans: B
Q26. The ideal time for giving oral premedication before surgery is: A. 15 minutes B. 30–60 minutes C. 60–90 minutes D. 2–3 hours	Ans: C
Q27. Which of the following systems must be evaluated during preoperative assessment? A. Only cardiovascular system B. Cardiovascular, respiratory, hepatic, renal, and nervous systems C. Only respiratory system D. Only the surgical site	Ans: B
Q28. A patient with controlled diabetes mellitus on oral hypoglycemics is classified as: A. ASA I B. ASA II C. ASA III D. ASA IV	Ans: B

Q29. A patient with unstable angina is classified as: A. ASA II B. ASA III C. ASA IV D. ASA V	Ans: C
Q30. Which preoperative investigation is essential before major surgery with expected blood loss? A. Echocardiography B. Blood grouping and cross-matching C. CT angiography D. MRI brain	Ans: B
Q31. Glycopyrrolate differs from atropine in that it: A. Crosses the blood-brain barrier B. Does not cross the blood-brain barrier C. Causes more tachycardia D. Has a longer duration of action on the heart	Ans: B
Q32. Which drug is used as premedication to prevent aspiration pneumonitis in high-risk patients? A. Midazolam B. Ranitidine + Metoclopramide + Sodium citrate C. Morphine D. Diazepam	Ans: B
Q33. Preoperative instructions regarding smoking cessation recommend stopping at least: A. 6 hours before surgery B. 24 hours before surgery C. 6–8 weeks before surgery D. 1 week before surgery	Ans: C
Q34. Which of the following is a contraindication to the use of atropine as premedication? A. Bradycardia B. Glaucoma C. Excessive secretions D. Vagal-mediated reflex	Ans: B
Q35. Which premedication drug has both antiemetic and anxiolytic properties? A. Glycopyrrolate B. Promethazine C. Ranitidine D. Sodium citrate	Ans: B

Q36. The MET (Metabolic Equivalent of Task) is used in preoperative assessment to evaluate: A. Airway difficulty B. Functional cardiac capacity C. Hepatic function D. Coagulation status	Ans: B
Q37. A patient who can climb two flights of stairs without symptoms has a functional capacity of at least: A. 1 MET B. 2 METs C. 4 METs D. 10 METs	Ans: C
Q38. Preoperative consent for anesthesia should include information about: A. Only the type of anesthesia B. Risks, benefits, alternatives, and plan of anesthesia C. Only the surgical procedure D. Only postoperative care	Ans: B
Q39. Which of the following oral hypoglycemic drugs should be stopped before surgery? A. Metformin B. Ranitidine C. Atenolol D. Amlodipine	Ans: A
Q40. Warfarin is typically stopped how many days before elective surgery? A. 1 day B. 2 days C. 3–5 days D. 10 days	Ans: C
Q41. In preoperative assessment, the term 'NPO' stands for: A. No Pain Observed B. Nil Per Oram (nothing by mouth) C. Normal Preoperative Order D. Non-Prescribed Opioid	Ans: B
Q42. The dose of midazolam commonly used for oral premedication in adults is: A. 0.1 mg B. 0.25–0.5 mg/kg C. 7.5–15 mg D. 50 mg	Ans: C

Q43. Which of the following is an advantage of midazolam over diazepam as premedication? A. Longer duration of action B. Water soluble, less painful on injection, and produces anterograde amnesia C. Greater respiratory depression D. Cheaper cost	Ans: B
Q44. Preoperative assessment of the cervical spine is important in patients with: A. Diabetes mellitus B. Rheumatoid arthritis and Down syndrome C. Hypertension D. Asthma	Ans: B
Q45. Which preoperative blood test is essential for all female patients of reproductive age? A. Thyroid function test B. Pregnancy test (urine/serum beta-hCG) C. Lipid profile D. Vitamin D levels	Ans: B
Q46. Preoperative instruction to patients about dentures includes: A. Keeping them in place during surgery B. Removing all dental prostheses before entering the operating room C. Gluing them in place D. Covering them with gauze	Ans: B
Q47. Dexmedetomidine used as premedication acts on: A. GABA receptors B. Alpha-2 adrenergic receptors C. Opioid receptors D. Dopamine receptors	Ans: B
Q48. Which of the following is true about premedication in pediatric patients? A. Anticholinergics are never used B. Midazolam syrup (0.5 mg/kg) is commonly used for anxiolysis C. Opioids are the first choice D. Premedication is not required	Ans: B
Q49. The fasting guideline for breast milk in infants before surgery is: A. 2 hours B. 4 hours C. 6 hours D. 8 hours	Ans: B

Q50. Which of the following investigations is indicated preoperatively in a patient with liver disease?

- A. ECG only
- B. Coagulation profile (PT, INR, aPTT)
- C. Chest X-ray only
- D. Blood sugar only

Ans: B

Answer Key & Explanations

Q1. Answer: B — To evaluate the patient's fitness for anesthesia and surgery

The primary purpose of preoperative assessment is to evaluate the patient's fitness for anesthesia and the planned surgical procedure.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q2. Answer: A — ASA I

ASA I denotes a normal healthy patient with no organic, physiological, or psychiatric disturbance.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q3. Answer: B — Mild systemic disease with no functional limitation

ASA II refers to a patient with mild systemic disease with no functional limitation (e.g., controlled hypertension, mild diabetes).

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q4. Answer: C — ASA V

ASA V is assigned to a moribund patient who is not expected to survive 24 hours with or without surgery.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q5. Answer: B — Emergency surgery

The suffix 'E' is added to denote an emergency surgical procedure.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q6. Answer: B — Airway and ease of intubation

Mallampati classification assesses the oropharyngeal structures to predict ease of intubation.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q7. Answer: B — Soft palate, fauces, uvula, and anterior and posterior tonsillar pillars

Mallampati Class I shows soft palate, fauces, uvula, and both tonsillar pillars fully visible.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q8. Answer: C — Hard palate only (soft palate not visible)

In Mallampati Class IV, only the hard palate is visible, indicating potentially difficult intubation.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q9. Answer: D — Ankle-brachial index

Ankle-brachial index is used for peripheral vascular disease assessment, not airway evaluation.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q10. Answer: B — Difficult intubation

A thyromental distance less than 6 cm (3 finger breadths) suggests potential difficulty in intubation.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q11. Answer: B — 2 hours

Current guidelines recommend fasting from clear fluids for at least 2 hours before elective surgery.

Topic: Instructions | Ref: Ch 4, Page 48

Q12. Answer: C — 6 hours

Patients should fast from solid food for at least 6 hours before elective surgery.

Topic: Instructions | Ref: Ch 4, Page 48

Q13. Answer: B — ECG

ECG is recommended as a routine investigation for all patients above 40 years of age.

Topic: Preoperative Assessment | Ref: Ch 4, Page 48

Q14. Answer: B — Antihypertensive medications

Antihypertensive medications should generally be continued on the morning of surgery to avoid rebound hypertension.

Topic: Instructions | Ref: Ch 4, Page 48

Q15. Answer: C — 7 days

Aspirin should be stopped 7 days before elective surgery as it irreversibly inhibits platelet function.

Topic: Instructions | Ref: Ch 4, Page 48

Q16. Answer: B — To allay anxiety and facilitate smooth induction

The primary goal of premedication is to allay anxiety, reduce secretions, and facilitate smooth induction of anesthesia.

Topic: Premedication | Ref: Ch 4, Page 50

Q17. Answer: B — Anxiolysis

Anxiolysis (relief of anxiety) is one of the primary desired effects of premedication.

Topic: Premedication | Ref: Ch 4, Page 50

Q18. Answer: B — Midazolam

Midazolam is the most commonly used benzodiazepine for premedication due to its rapid onset, short duration, and amnestic properties.

Topic: Premedication | Ref: Ch 4, Page 50

Q19. Answer: B — Glycopyrrolate

Glycopyrrolate is an anticholinergic agent used as a premedicant primarily for its antisialagogue (drying) effect.

Topic: Premedication | Ref: Ch 4, Page 50

Q20. Answer: B — Atropine

Atropine is a tertiary amine that crosses the blood-brain barrier and can cause central anticholinergic effects.

Topic: Premedication | Ref: Ch 4, Page 50

Q21. Answer: C — Induction of surgical anesthesia

Induction of surgical anesthesia is not a goal of premedication; it is achieved by induction agents.

Topic: Premedication | Ref: Ch 4, Page 50

Q22. Answer: B — Blocking H2 receptors to reduce gastric acid secretion

Ranitidine is an H2 receptor antagonist that reduces gastric acid volume and increases pH.

Topic: Premedication | Ref: Ch 4, Page 50

Q23. Answer: B — Increasing gastric emptying and increasing lower esophageal sphincter tone

Metoclopramide promotes gastric emptying and increases lower esophageal sphincter tone, reducing aspiration risk.

Topic: Premedication | Ref: Ch 4, Page 50

Q24. Answer: B — Neutralize gastric acidity

Sodium citrate (0.3 M) is a non-particulate antacid used to neutralize gastric acid and raise gastric pH.

Topic: Premedication | Ref: Ch 4, Page 50

Q25. Answer: B — Morphine

Morphine is used as a premedicant for analgesia and sedation, especially in patients with pain.

Topic: Premedication | Ref: Ch 4, Page 50

Q26. Answer: C — 60–90 minutes

Oral premedication is generally given 60–90 minutes before surgery for adequate absorption and effect.

Topic: Premedication | Ref: Ch 4, Page 50

Q27. Answer: B — Cardiovascular, respiratory, hepatic, renal, and nervous systems

A thorough preoperative assessment evaluates all major organ systems including cardiovascular, respiratory, hepatic, renal, and nervous systems.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q28. Answer: B — ASA II

A patient with well-controlled systemic disease like mild diabetes is classified as ASA II.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q29. Answer: C — ASA IV

Unstable angina represents a severe systemic disease that is a constant threat to life, hence ASA IV.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q30. Answer: B — Blood grouping and cross-matching

Blood grouping and cross-matching is essential before any major surgery where significant blood loss is anticipated.

Topic: Preoperative Assessment | Ref: Ch 4, Page 48

Q31. Answer: B — Does not cross the blood-brain barrier

Glycopyrrolate is a quaternary ammonium compound that does not cross the blood-brain barrier, hence no central effects.

Topic: Premedication | Ref: Ch 4, Page 50

Q32. Answer: B — Ranitidine + Metoclopramide + Sodium citrate

The combination of H2 blocker, prokinetic, and non-particulate antacid is used to reduce aspiration risk.

Topic: Premedication | Ref: Ch 4, Page 50

Q33. Answer: C — 6–8 weeks before surgery

Ideally, smoking should be stopped 6–8 weeks before surgery to allow improvement in mucociliary function and reduction in secretions.

Topic: Instructions | Ref: Ch 4, Page 48

Q34. Answer: B — Glaucoma

Atropine is contraindicated in patients with narrow-angle glaucoma as it can precipitate an acute attack.

Topic: Premedication | Ref: Ch 4, Page 50

Q35. Answer: B — Promethazine

Promethazine (a phenothiazine) has antiemetic, sedative, and mild anxiolytic properties making it useful as premedication.

Topic: Premedication | Ref: Ch 4, Page 50

Q36. Answer: B — Functional cardiac capacity

METs are used to assess a patient's functional cardiac capacity and exercise tolerance preoperatively.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q37. Answer: C — 4 METs

Climbing two flights of stairs corresponds to approximately 4 METs, which is considered adequate for most surgeries.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q38. Answer: B — Risks, benefits, alternatives, and plan of anesthesia

Informed consent for anesthesia must cover the anesthetic plan, risks, benefits, and available alternatives.

Topic: Instructions | Ref: Ch 4, Page 48

Q39. Answer: A — Metformin

Metformin should be withheld before surgery due to the risk of lactic acidosis, especially if renal function is compromised.

Topic: Instructions | Ref: Ch 4, Page 48

Q40. Answer: C — 3–5 days

Warfarin is typically stopped 3–5 days before elective surgery to allow INR to normalize.

Topic: Instructions | Ref: Ch 4, Page 48

Q41. Answer: B — Nil Per Oram (nothing by mouth)

NPO stands for Nil Per Oram, meaning nothing by mouth, referring to preoperative fasting.

Topic: Instructions | Ref: Ch 4, Page 48

Q42. Answer: C — 7.5–15 mg

Oral midazolam 7.5–15 mg is commonly used for adult premedication.

Topic: Premedication | Ref: Ch 4, Page 50

Q43. Answer: B — Water soluble, less painful on injection, and produces anterograde amnesia

Midazolam is water-soluble (less painful on IV injection), has a shorter duration, and provides reliable anterograde amnesia.

Topic: Premedication | Ref: Ch 4, Page 50

Q44. Answer: B — Rheumatoid arthritis and Down syndrome

Patients with rheumatoid arthritis and Down syndrome may have atlantoaxial instability requiring cervical spine assessment.

Topic: Preoperative Assessment | Ref: Ch 4, Page 47

Q45. Answer: B — Pregnancy test (urine/serum beta-hCG)

A pregnancy test should be considered for all female patients of childbearing age before elective surgery.

Topic: Preoperative Assessment | Ref: Ch 4, Page 48

Q46. Answer: B — Removing all dental prostheses before entering the operating room

All removable dental prostheses should be removed preoperatively to prevent airway obstruction or damage during intubation.

Topic: Instructions | Ref: Ch 4, Page 48

Q47. Answer: B — Alpha-2 adrenergic receptors

Dexmedetomidine is a selective alpha-2 adrenergic agonist that provides sedation, anxiolysis, and analgesia.

Topic: Premedication | Ref: Ch 4, Page 50

Q48. Answer: B — Midazolam syrup (0.5 mg/kg) is commonly used for anxiolysis

Oral midazolam (0.5 mg/kg) is the most commonly used premedicant in children for anxiolysis and separation from parents.

Topic: Premedication | Ref: Ch 4, Page 50

Q49. Answer: B — 4 hours

Breast milk has a fasting time of 4 hours before elective surgery in infants.

Topic: Instructions | Ref: Ch 4, Page 48

Q50. Answer: B — Coagulation profile (PT, INR, aPTT)

Coagulation profile is essential in liver disease patients as the liver synthesizes most clotting factors.

Topic: Preoperative Assessment | Ref: Ch 4, Page 48